

Apache Cassandra Implementation for Music Streaming Data Management

Objective

The objective of this project is to install and configure Apache Cassandra on Ubuntu WSL, create a NoSQL database for a music streaming application, and design optimized tables to efficiently answer application-specific queries. The project demonstrates Cassandra data modeling, table creation, data insertion, and query execution using CQL.

Project Description

This project uses **Apache Cassandra**, a distributed NoSQL database, to build a database for the **Apache Cassandra Implementation for Music Streaming Data Management**.

Initially, Cassandra is installed using Docker (recommended for WSL) or through a direct Ubuntu installation. A keyspace named **sparkify** is created, followed by three tables designed specifically to support different query patterns:

- Retrieve all songs played in a specific session.
- Retrieve all songs played by a user during a session in reverse chronological order.
- Find all users who listened to a particular song.

Sample data is inserted into each table, and CQL queries are executed to verify the database design and retrieve the required information. The project also includes optional Python-based data generation for testing with larger datasets and demonstrates Cassandra features such as UPSERT operations, partition keys, clustering order, and consistency levels.

Screenshots

```
pratham@LAPTOP-PL4G895R:~$ sudo apt install -y \
apt-transport-https \
ca-certificates \
curl \
gnupg \
lsb-release
apt-transport-https is already the newest version (3.2.0).
ca-certificates is already the newest version (20260601~26.04.1).
curl is already the newest version (8.18.0-1ubuntu2.2).
gnupg is already the newest version (2.4.8-4ubuntu3).
lsb-release is already the newest version (12.1-2build1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
pratham@LAPTOP-PL4G895R:~$
```

```
pratham@LAPTOP-PL4G895R:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQINBFit2ioBEADhWpZ8/wvZ6hUTiXOwQHxMAlaFHCpH9hAtr4F1y2+OYdbtMuth
lqwp028AqY+PRfVMtSYMbjuQuu5byyKR01BbqYhuS3jtqQmLjZ/bJvXqnmIVXh
38UuLa+z077PxyxQhu58BqntTPQMfiiyqEiU+BKbq2WmANUKQf+1AmZY/IruOXbnq
L4C1+gJ8vfmXQt99npCaxEjaNRVYf0S8QcixNzHUYNb6emjLAnyEVLZzeqo7XKl7
UrwV5inawTSzWNvtjEjj4nJL8NsLwscPLQQUhTQ+7BbQXAwAmeHCUTQIvvWXqW0N
cmhh4HgeQscQHYgOJjJdVfoY5MucvgLbIgCqfzAHW9jxmRL4qbMZj+b1XoePEtht
ku4bIQN1X5P07fNwZlgaRL5Z4POXDDZTLIQ/EL58j9kp4bnWRCJW0lya+f8ocodo
vZZ+Doi+fy4D5ZGrL4XEcIQP/Lv5uFyf+kQtL/94VFYVJ0leAv8W92KdgDkhTcTD
G7c0tIkVEKNUq48b3aQ64N0ZQW7fVjfoKwEZd0qPE72Pa45jrZzvUFxSpdiNk2tZ
XYukHjLxxEgBdC/J3cMMNRE1F4NCA3ApfV1Y7/hTeOnmDuDYwr9/obA8t016Vlj
q5rdkymPf4Jf8mXUW5eCN1vAFHxeg9ZWemhBtQmGxXnw9M+z6hWwc6ahmwARAQAB
tCtEb2NrZXIgmUvS2WfZSSAoQ0UgZGVhKSA8ZG9ja2VvQGRvY2tLc15jb20+iQI3
BBMBCgAhBQJYrefAAhsVbQsJCAcDBRUKCQgLBRYCAwEAAh4BAheAAAJE12BgDw0
v82IsskP/iQZo68fLDQmNvn8X5XTd6RRaUH33kXYXquT6NkHJcIS7E2gTJmqvMqd
tI4mNYHCSEYxISgrcYV5YqX9P6+Ko+vozo4nseUQLPH/ATQ4qL0Zok+1jkag3Lgk
jonyUf9bwtwXfP05HC3GMHPPhhcUSexCxQLQvnFWXD2sWLKivHp2fT8QbRGeZ+d3m
6fqcd5Fu7pxsqm0EUDK5NL+nPIgYhN+auTrhgzhK1CSHfGccM/wFRlei9Utz6p9P
XRKILWnXt4qNGZNTN0tR+NLG/6Bqd80YBaFAUcuc/w1VW6JQ2VGYZHnZu9S8LMc
FYBa5Ig9PxxwGQ0gq6RDkDbv+PqTQT5EFMeR1mrjckk4DQJjbxemZbiNMG5kGECA8
g383P3eLhn03WGbEa4MNC3Z4+7c236QI3xWJfNPdUbXRaAwHy/6rTSFbzwKB0Jm
ebwzQfwjQY6f55MiI/RqDCyuPj3r3jyVRkK86pQKBAJwFHyqj9KaKXmZjfvNwLh
9svIGfNbGHpucATqREvUHuQbNnqkCx8VvhtYkhDb9fEP2xBu5VvHbR+3nfVhMut5
G34Ct5RS7Jt6LlFfDctn8CaSas/L1HbiGeRgc70X/9aYx/V/CEJv0lE8gP6uDow
FPIZ7d6vH+Vro6xuwEGiuMaiznap2KhZmpkgfupyFmplh0s6knyuQINBFit2ioB
EADneL9S9m4vhu3bLaRjVUUYj7b/qTjCylvCH5XUE6R2k+ckEzjfAMZPLp0+/tF
M2JJMD4SiFkuS3xck9KtZGcufGmcwILQRzeHF7vJUKrLD5RTkNi23ydvWZgPjtx
Q+DTT1Zcn7BrQFY6FgnRoUVIxtwdw1bMY/89rsFgS5wuMESd3Q2RYgb7E0F0pnu
w6da7WakWf4IhF5nsNYGDVAIHZpiqCL+uTbf1epCjr0LiZkZ3Z3Yk5CM/TiFzPk
z2LLZ89cpD8U+NtCsFagWwfjd2U3jDapgH+7nQnCEwPR0tzaKHG6LA3pXdx5zG8
eRc6/0ibU5WvfjKxLLPfNecS2pCL3IEI5nothEEYDQH6szpLog79xB9dVnJyKJb
VfxXnseoYqVrZr2VVbUI5B1wm6B40E3eGVfUQWiuX54DspyVMMk41Mx7QJ3iynIa
1N4ZAqVMAEruxYXTRTxc9XW0tYhDMA/1GYvz0EmFpm8LzTHA6sFvTpm/ZLNCX6P1X
zJwrv7DSQKD6GGLBQUX+OeEJ8tKkfbQOTJSPUDh8P8YxDfS5E0GAvhmpBYD42kQ
pqXjEC+XcycTvGI7impv9PDY1RCC1zkBjKPa120rNhv/hkVh/YhuGoaJoHyy4h7
ZQopdcMtP2dgmhEegny9JCSwxFQmQ0zK0g7m6SHiKmwjwARAQABiQQ+BBgBCAAJ
BQJYrdogAhsCAiKJIEI2BgDw0v82IwV0gBBkBCAAGBQJYrdogAAoJEH6gqcPyc/zY
1WAP/2wJ+R0gE6qsc3rjaIz58PJmc8goKrir5hnELWhPgbq7cYIsW5qiFyLhkdp
YcMmhD9mRiPpQn6Ya2w3e3B8zfIVKipbMBNke/ytZ9M7qHmDCCjoismwEXN3wKYI
mD9VHONsL/CG1rU9Isw1jtB5g1YxuBA7M/m36XN6x2u+NtNMDB9P56yc4gfsZVES
KA9v+yY2/L45L8d/WUKU10YXomn6hyBGI7JrBLq0CX37GEYP609rrKipfz73Xf07
JIGz0KZLLjb/D9RX/g7nRbCn+3EtH7xnk+TK/50euEKw8SMUg147sJTCpQmv6UzZ
cM4JgLOhBHVCojv4C/plELwMddAL0FeYQzTif6sMRPf+3DSj8frbInjCh3cy0LY0
6br92Kfom17EIJ2CACoeq7UPhi2oouYBwPxx5ytdelhKjoo+sN7RIWua6P2WsmoN5
U888cSylXC0+ADFDgLX9K2zrDVYUG1vo8CX0vzxFBAHmN6Px26fHIT1/hYUHQRIz
VfNDcyQmXqkOnZvvoMfz/Q0s9BhFJ/zU6AgQbIZE/hm1spsfgvtsD1frZfygXJ9f
irP+MSAI80xHSF91qSRZOj4P13ZJNbq4Yxv0b1pkMgeGdjdCYhLU+LZ4wbQmpCk
Sve2prLLureigXtmZfKqevRz7FrIZiu9ky8wnCAPwC7/zmS18rgP/17b0tL4/iIz
QhxAAoAMWVrGyJivSkjSGxluCojsWfsTAm11P7jsruIL61ZzMUVE2aM3Pmj5G+W
9AcZ58Em+1WsVnAXdUR//bMmhyr8wL/G1Y01V3JEJTRdxsSxdYa4deGBBY/Adpsw
24jxh0JR+lsJppIUeb999+R8eudhRHG9eF07DRu6weatUJ6suupoDTRWtr/4yGqe
dKxV3qQhNLSnaAzqW/1nA3iUB4k7kCaKZxhdhDbCLf9P37qaRW467BLCVO/coL3y
Vm50dwdrNtKpMBh3ZpbB1uJvgi9mXtyBOMJ3v8RZeDzFiG8HdCtg9RvIt/AIFoHR
H3S+U79NT6i0KPzLImDfs8T7RlpyuMc4Ufs8ggyg9v3Ae6cN3eQyxck3w0cbBwsh
/nQNFsA6uu+9H7NhbhBMhYnpNZyrHzCmzyXkauwRAqoCbGCNykTRwsur9gS41TQ
M8ssD1jFheOJf3h0DnkKU+HKjvMR0L1DK7zdmLDnZA1cvtZH/nCC9KPj1z8QC47S
xx+dTZSx40NAHwbs/LN3PoKtn8LPjY9NP9uDWI+TWYquS2U+KHDrBDLsgozDbs/O
jCxcPdzNmXpWQHETHU76490XHP7UeNST1mCUC5qdank0V1ieJf6/CFTFU4MfcrG
YT90qFF93M3v01BbxP+EIY2/9tiIPbrd
=0YYh
-----END PGP PUBLIC KEY BLOCK-----
```

```
pratham@LAPTOP-PL4G895R:~$ echo \
> "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-archive-
keyring.gpg] https://download.docker.com/linux/ubuntu \
> $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
pratham@LAPTOP-PL4G895R:~$ sudo usermod -aG docker $pratham
Usage: usermod [options] LOGIN
```

Options:

-a, --append	append the user to the supplemental GROUPS mentioned by the -G option without removing the user from other groups
-b, --badname	allow bad names (DEPRECATED)
-c, --comment COMMENT	new value of the GECOS field
-d, --home HOME_DIR	new home directory for the user account
-e, --expiredate EXPIRE_DATE	set account expiration date to EXPIRE_DATE
-f, --inactive INACTIVE	set password inactive after expiration to INACTIVE
-g, --gid GROUP	force use GROUP as new primary group
-G, --groups GROUPS	new list of supplementary GROUPS
-h, --help	display this help message and exit
-l, --login NEW_LOGIN	new value of the login name
-L, --lock	lock the user account
-m, --move-home	move contents of the home directory to the new location (use only with -d)
-o, --non-unique	allow using duplicate (non-unique) UID
-p, --password PASSWORD	use encrypted password for the new password
-P, --prefix PREFIX_DIR	prefix directory where are located the /etc/* files
-r, --remove	remove the user from only the supplemental GROUPS mentioned by the -G option without removing the user from other groups
-R, --root CHROOT_DIR	directory to chroot into
-s, --shell SHELL	new login shell for the user account
-u, --uid UID	new UID for the user account
-U, --unlock	unlock the user account
-v, --add-subuids FIRST-LAST	add range of subordinate uids
-V, --del-subuids FIRST-LAST	remove range of subordinate uids
-w, --add-subgids FIRST-LAST	add range of subordinate gids
-W, --del-subgids FIRST-LAST	remove range of subordinate gids
-Z, --selinux-user SEUSER	new SELinux user mapping for the user account
--selinux-range SERANGE	new SELinux MLS range for the user account

```
pratham@LAPTOP-PL4G895R:~$ mkdir -p ~/cassandra-exercise
```

```
pratham@LAPTOP-PL4G895R:~$ cd ~/cassandra-exercise
```

```
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ cat > docker-compose.yml << 'EOF'
```

```
> services:
```

```
cassandra:
```

```
image: cassandra:latest
```

```
container_name: cassandra
```

```
hostname: cassandra
```

```
ports:
```

```
- "9042:9042"
```

```
- "9160:9160"
```

```
volumes:
```

```
- "./cassandra-data:/var/lib/cassandra"
```

```
- "./scripts:/scripts"
```

```
environment:
```

```
- CASSANDRA_CLUSTER_NAME=MyCluster
```

```
- CASSANDRA_DC=datacenter1
```

```
- CASSANDRA_RACK=rack1
```

```
- CASSANDRA_ENDPOINT_SNITCH=GossipingPropertyFileSnitch
```

```
- CASSANDRA_NUM_TOKENS=256
```

```
- MAX_HEAP_SIZE=512M
```

```
- HEAP_NEWSIZE=100M
```

```
restart: unless-stopped
```

```
EOF
```

```
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ echo "♦♦♦♦♦ Waiting for Cassandra to start (60 seconds)..."
♦♦♦♦♦ Waiting for Cassandra to start (60 seconds)...
```

```
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ sudo docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
7456a8195ab2   cassandra:latest "docker-entrypoint.s..." 17 seconds ago Up 15 seconds 7000-7001/tcp, 7199/tcp, 9160/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ sudo docker logs cassandra | tail -20
WARNING: A terminally deprecated method in java.lang.System has been called
WARNING: System:setSecurityManager has been called by org.apache.cassandra.security.ThreadAwareSecurityManager (file:/opt/cassandra/lib/apache-cassandra-5.0.8.jar)
WARNING: Please consider reporting this to the maintainers of org.apache.cassandra.security.ThreadAwareSecurityManager
WARNING: System:setSecurityManager will be removed in a future release
INFO [main] 2026-07-09T03:26:45,264 LocalSessions.java:416 - LocalSessions start completed in 2 ms, sessions loaded from DB: 0
INFO [main] 2026-07-09T03:26:45,266 StreamManager.java:267 - Storing streaming state for 3d or for size 41943040
INFO [main] 2026-07-09T03:26:45,267 QueryProcessor.java:198 - Preloaded 0 prepared statements in 0 ms
INFO [main] 2026-07-09T03:26:45,267 StorageService.java:968 - Cassandra version: 5.0.8
INFO [main] 2026-07-09T03:26:45,267 StorageService.java:969 - Cit SMA: 172227094626d50d4bc9a3d9de2e3659a7844386
INFO [main] 2026-07-09T03:26:45,267 StorageService.java:970 - CQL version: 3.4.7
INFO [main] 2026-07-09T03:26:45,267 StorageService.java:971 - Native protocol supported versions: 3/v3, 4/v4, 5/v5, 6/v6-beta (default: 5/v5)
INFO [main] 2026-07-09T03:26:45,289 IndexSummaryManager.java:105 - Initializing index summary manager with a memory pool size of 94 MB and a resize interval of 60 minute
s
INFO [main] 2026-07-09T03:26:45,298 StorageService.java:988 - Loading persisted ring state
INFO [main] 2026-07-09T03:26:45,298 StorageService.java:1886 - Populating token metadata from system tables
INFO [main] 2026-07-09T03:26:45,305 InboundConnectionInitiator.java:164 - Listening on address: (/172.17.0.2:7000), nic: eth0, encryption: unencrypted
INFO [main] 2026-07-09T03:26:45,379 StorageService.java:880 - Unable to gossip with any peers but continuing anyway since node is in its own seed list
INFO [main] 2026-07-09T03:26:45,430 StorageService.java:1216 - Starting up server gossip
INFO [main] 2026-07-09T03:26:45,433 ColumnFamilyStore.java:1052 - Enqueuing flush of system.local, Reason: INTERNALLY_FORCED, Usage: 525B (0%) on-heap, 0B (0%) off-heap
INFO [PerDiskMemtableFlushWriter:0:1] 2026-07-09T03:26:45,442 Flushing.java:153 - Writing Memtable-local8958428162(538 serialized bytes, 1 ops, 525B (0%) on-heap, 0B (0%) off-heap), flushed range = (null, null)
INFO [PerDiskMemtableFlushWriter:0:1] 2026-07-09T03:26:45,443 Flushing.java:179 - Completed flushing /var/lib/cassandra/data/system/local-7ad54392bcd35a684174e847860b377/nb-5-big-Data.db (298) for commitlog position CommitLogPosition(segmentId=1783567603534, position=863)
INFO [MemtableFlushWriter:2] 2026-07-09T03:26:45,470 LogTransaction.java:249 - Unfinished transaction log, deleting /var/lib/cassandra/data/system/local-7ad54392bcd35a684174e847860b377/nb_txn_flush.030d81a0-7b46-11f1-acac-39de909c56e.log
INFO [main] 2026-07-09T03:26:45,553 MigrationCoordinator.java:269 - Starting migration coordinator and scheduling pulling schema versions every PT1M
INFO [main] 2026-07-09T03:26:45,564 SnapshotManager.java:135 - Scheduling expired snapshot cleanup with initialDelaySeconds=5 and cleanupPeriodSeconds=60
INFO [main] 2026-07-09T03:26:45,644 StorageService.java:1294 - This node will not auto bootstrap because it is configured to be a seed node.
```

```
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ sudo apt install python3-venv -y
Installing:
  python3-venv

Installing dependencies:
  python3-pip-whl python3-setuptools-whl python3.14-venv

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 2
  Download size: 2610 kB
  Space needed: 2958 kB / 1018 GB available

Get:1 http://archive.ubuntu.com/ubuntu/resolute/universe amd64 python3-pip-whl all 25.1.1+dfsg-1ubuntu2 [1430 kB]
Get:2 http://archive.ubuntu.com/ubuntu/resolute/universe amd64 python3-setuptools-whl all 78.1.1-0.1build1 [1174 kB]
Get:3 http://archive.ubuntu.com/ubuntu/resolute-updates/universe amd64 python3.14-venv amd64 3.14.4-1ubuntu0.1 [5354 B]
Get:4 http://archive.ubuntu.com/ubuntu/resolute/universe amd64 python3-venv amd64 3.14.3-0ubuntu2 [1032 B]
Fetched 2610 kB in 5s (557 kB/s)
Selecting previously unselected package python3-pip-whl.
(Reading database ... 49654 files and directories currently installed.)
Preparing to unpack .../python3-pip-whl_25.1.1+dfsg-1ubuntu2_all.deb ...
Unpacking python3-pip-whl (25.1.1+dfsg-1ubuntu2) ...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack .../python3-setuptools-whl_78.1.1-0.1build1_all.deb ...
Unpacking python3-setuptools-whl (78.1.1-0.1build1) ...
Selecting previously unselected package python3.14-venv.
Preparing to unpack .../python3.14-venv_3.14.4-1ubuntu0.1_amd64.deb ...
Unpacking python3.14-venv (3.14.4-1ubuntu0.1) ...
Selecting previously unselected package python3-venv.
Preparing to unpack .../python3-venv_3.14.3-0ubuntu2_amd64.deb ...
Unpacking python3-venv (3.14.3-0ubuntu2) ...
Setting up python3-setuptools-whl (78.1.1-0.1build1) ...
Setting up python3-pip-whl (25.1.1+dfsg-1ubuntu2) ...
Setting up python3.14-venv (3.14.4-1ubuntu0.1) ...
Setting up python3-venv (3.14.3-0ubuntu2) ...
```

```
pratham@LAPTOP-PL4G895R:~$ cd ~/cassandra-exercise
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ nano setup_exercise.cql
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ sudo docker cp setup_exercise.cql cassandra:/setup_exercise.cql
[sudo: authenticate] Password:
Successfully copied 3.41kB (transferred 5.12kB) to cassandra:/setup_exercise.cql
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ sudo docker exec -it cassandra cqlsh -f /setup_exercise.cql
/setup_exercise.cql:26:SyntaxException: line 1:0 no viable alternative at input 'session_id' ([session_id]...)
/setup_exercise.cql:77:SyntaxException: line 1:58 mismatched input ' '; expecting K_FROM (SELECT '권권권권권 TABLES CREATED AND DATA INSERTED!' as status[;])

song_info_by_session song_playing_history_by_user who_listened_to_song

session_id | item_in_session | artist | length | song
-----|-----|-----|-----|-----
100 | 1 | The Beatles | 7.11 | Hey Jude
100 | 2 | Led Zeppelin | 8.02 | Stairway to Heaven
100 | 3 | Pink Floyd | 6.53 | Comfortably Numb
101 | 1 | Queen | 5.55 | Bohemian Rhapsody
(4 rows)

user_id | session_id | item_in_session | artist | first_name | last_name | song
-----|-----|-----|-----|-----|-----|-----
2 | 200 | 1 | Queen | Jane | Smith | Bohemian Rhapsody
1 | 100 | 3 | Pink Floyd | John | Doe | Comfortably Numb
1 | 100 | 2 | Led Zeppelin | John | Doe | Stairway to Heaven
1 | 100 | 1 | The Beatles | John | Doe | Hey Jude
(4 rows)

song | user_id | first_name | last_name
-----|-----|-----|-----
Comfortably Numb | 1 | John | Doe
Hey Jude | 1 | John | Doe
Hey Jude | 2 | Jane | Smith
(3 rows)
```

```
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ sudo docker exec -it cassandra cqlsh
ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> DESC KEYSPACES;

sparkify      system_auth      system_traces
student_db    system_distributed  system_views
system        system_schema      system_virtual_schema

cqlsh> USE sparkify;
cqlsh:sparkify> DESC TABLES;

song_info_by_session  song_playing_history_by_user  who_listened_to_song

cqlsh:sparkify> SELECT * FROM song_info_by_session
... WHERE session_id = 100;

session_id | item_in_session | artist      | length | song
-----+-----+-----+-----+-----
100        | 1               | The Beatles | 7.11   | Hey Jude
100        | 2               | Led Zeppelin | 8.02   | Stairway to Heaven
100        | 3               | Pink Floyd  | 6.53   | Comfortably Numb

(3 rows)
```

```
cqlsh:sparkify> SELECT *
... FROM song_playing_history_by_user
... WHERE user_id = 1
... AND session_id = 100;

user_id | session_id | item_in_session | artist      | first_name | last_name | song
-----+-----+-----+-----+-----+-----+-----
1        | 100        | 3               | Pink Floyd  | John       | Doe       | Comfortably Numb
1        | 100        | 2               | Led Zeppelin | John       | Doe       | Stairway to Heaven
1        | 100        | 1               | The Beatles | John       | Doe       | Hey Jude

(3 rows)
cqlsh:sparkify> SELECT *
... FROM who_listened_to_song
... WHERE song = 'Hey Jude';

song      | user_id | first_name | last_name
-----+-----+-----+-----
Hey Jude  | 1       | John      | Doe
Hey Jude  | 2       | Jane      | Smith

(2 rows)
cqlsh:sparkify> INSERT INTO song_info_by_session
... (session_id, item_in_session, artist, song, length)
... VALUES
... (102, 1, 'Nirvana', 'Smells Like Teen Spirit', 5.01);
cqlsh:sparkify> INSERT INTO song_info_by_session
... (session_id, item_in_session, artist, song, length)
... VALUES
... (100, 1, 'The Beatles', 'Let It Be', 4.03);
cqlsh:sparkify> SELECT *
... FROM song_info_by_session
... WHERE session_id = 100;

session_id | item_in_session | artist      | length | song
-----+-----+-----+-----+-----
100        | 1               | The Beatles | 4.03   | Let It Be
100        | 2               | Led Zeppelin | 8.02   | Stairway to Heaven
100        | 3               | Pink Floyd  | 6.53   | Comfortably Numb

(3 rows)
```

```
cqlsh:sparkify> CONSISTENCY;
Current consistency level is ONE.
cqlsh:sparkify> CONSISTENCY ONE;
Consistency level set to ONE.
cqlsh:sparkify> SELECT *
... FROM song_info_by_session
... WHERE session_id = 100;
```

session_id	item_in_session	artist	length	song
100	1	The Beatles	4.03	Let It Be
100	2	Led Zeppelin	8.02	Stairway to Heaven
100	3	Pink Floyd	6.53	Comfortably Numb

(3 rows)

```
cqlsh:sparkify> EXIT;
```

```
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ cd ~/cassandra-exercise
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ sudo apt install python3-venv -y
python3-venv is already the newest version (3.14.3-0ubuntu2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ python3 -m venv venv
pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ source venv/bin/activate
(venv) pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ (venv) pratham@LAPTOP-PL4G895R:~/cassandra-exercise$
-bash: syntax error near unexpected token `pratham@LAPTOP-PL4G895R:~/cassandra-exercise$'
(venv) pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ pip install cassandra-driver
Collecting cassandra-driver
  Downloading cassandra_driver-3.30.1-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (6.8 kB)
Collecting geomet>=1.1 (from cassandra-driver)
  Downloading geomet-1.1.0-py3-none-any.whl.metadata (11 kB)
Collecting Deprecated>=1.3.1 (from cassandra-driver)
  Downloading deprecated-1.3.1-py2.py3-none-any.whl.metadata (5.9 kB)
Collecting wrapt<3,>=1.10 (from Deprecated>=1.3.1->cassandra-driver)
  Downloading wrapt-2.2.2-cp314-cp314-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (7.4 kB)
Collecting click (from geomet>=1.1->cassandra-driver)
  Downloading click-8.4.2-py3-none-any.whl.metadata (2.6 kB)
Downloading cassandra_driver-3.30.1-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (3.5 MB)
3.5/3.5 MB 116.7 kB/s eta 0:00:00
Downloading deprecated-1.3.1-py2.py3-none-any.whl (11 kB)
Downloading wrapt-2.2.2-cp314-cp314-manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (167 kB)
Downloading geomet-1.1.0-py3-none-any.whl (31 kB)
Downloading click-8.4.2-py3-none-any.whl (119 kB)
Installing collected packages: wrapt, click, geomet, Deprecated, cassandra-driver
Successfully installed Deprecated-1.3.1 cassandra-driver-3.30.1 click-8.4.2 geomet-1.1.0 wrapt-2.2.2
(venv) pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ python -c "from cassandra.cluster import Cluster; print('Driver Installed Successfully')"
```

```
(venv) pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ nano generate_data.py
(venv) pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ source venv/bin/activate
(venv) pratham@LAPTOP-PL4G895R:~/cassandra-exercise$ python generate_data.py
100 Random Records Inserted Successfully!
```

```
(venv) pratham@LAPTOP-PLNG895R:~/cassandra-exercise$ sudo docker exec -it cassandra cqlsh
```

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See <https://cassandra.apache.org/doc/latest/tools/cqlsh.html> for more information.

```
Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> USE sparkify;
cqlsh:sparkify> SELECT * FROM song_info_by_session;
```

session_id	item_in_session	artist	length	song
363	1	U2	5.56	Hey Jude
363	2	Queen	3.18	Hotel California
363	3	Mirvana	3.1	Smells Like Teen Spirit
363	4	Queen	4.11	With or Without You
363	5	Mirvana	4.54	Hey Jude
363	6	Queen	2.73	Smells Like Teen Spirit
363	7	Led Zeppelin	7.94	Back in Black
144	1	Michael Jackson	3.63	Comfortably Numb
144	2	Michael Jackson	3.56	With or Without You
144	3	AC/DC	9.95	Smells Like Teen Spirit
144	4	U2	9.24	Like a Prayer
144	5	Led Zeppelin	5.16	Smells Like Teen Spirit
144	6	AC/DC	8.65	With or Without You
144	7	Eagles	5.4	Hotel California
128	1	Queen	9.8	With or Without You
781	1	Madonna	7.26	Comfortably Numb
781	2	Pink Floyd	6.39	Bohemian Rhapsody
781	3	Pink Floyd	5.57	Like a Prayer
781	4	U2	5.87	Stairway to Heaven
781	5	Eagles	9.38	Comfortably Numb
781	6	Pink Floyd	5.62	Like a Prayer
781	7	U2	4.8	Hotel California
857	1	Eagles	4.64	Billie Jean
857	2	AC/DC	3.8	Stairway to Heaven
857	3	U2	9.1	Back in Black
857	4	Queen	4.71	Back in Black
857	5	Eagles	6.35	With or Without You
857	6	Mirvana	2.77	Stairway to Heaven
857	7	Pink Floyd	8.81	Hey Jude
857	8	Eagles	6.75	Hotel California
818	1	Led Zeppelin	8.53	Like a Prayer
818	2	U2	3.05	Billie Jean
818	3	Led Zeppelin	3.35	Like a Prayer
818	4	Madonna	8.78	Stairway to Heaven
514	1	Eagles	6.8	Hey Jude
514	2	Pink Floyd	4.74	Comfortably Numb
514	3	Led Zeppelin	8.14	Like a Prayer
377	1	Eagles	3.95	With or Without You
377	2	U2	3.75	Hey Jude
377	3	Eagles	3.79	Back in Black
377	4	Eagles	7.38	Stairway to Heaven
377	5	Led Zeppelin	6.8	Like a Prayer
377	6	AC/DC	5.15	Stairway to Heaven
377	7	Queen	2.52	Back in Black
377	8	Led Zeppelin	6.25	Smells Like Teen Spirit
377	9	Eagles	9.49	Billie Jean
965	1	The Beatles	4.85	Bohemian Rhapsody
245	1	The Beatles	4.58	Hotel California
245	2	Pink Floyd	8.88	Comfortably Numb
245	3	Madonna	7.17	Back in Black
245	4	Mirvana	4.29	Bohemian Rhapsody
245	5	Mirvana	4.2	Bohemian Rhapsody
245	6	The Beatles	9.31	With or Without You
361	1	Madonna	9.14	Back in Black
361	2	Pink Floyd	5.89	Smells Like Teen Spirit
361	3	Madonna	9.95	Hey Jude
361	4	The Beatles	4.75	Stairway to Heaven
456	1	Mirvana	5.27	Hey Jude
456	2	Pink Floyd	9.43	Bohemian Rhapsody
456	3	Led Zeppelin	6.61	Smells Like Teen Spirit
456	4	Michael Jackson	6.72	With or Without You
456	5	Queen	3.61	Bohemian Rhapsody
456	6	Eagles	9.08	Back in Black
456	7	Eagles	5.39	With or Without You
828	1	Madonna	8.4	Hotel California
828	2	The Beatles	6.62	Smells Like Teen Spirit
828	3	The Beatles	4.95	Like a Prayer
828	4	Queen	3.68	Hey Jude
946	1	Led Zeppelin	6.81	Smells Like Teen Spirit
875	1	AC/DC	6.57	Billie Jean
875	2	AC/DC	5.94	Hotel California
875	3	Led Zeppelin	4.43	Like a Prayer
875	4	Led Zeppelin	5.14	Like a Prayer
875	5	U2	6.55	Comfortably Numb
875	6	Eagles	8.88	Back in Black
875	7	Eagles	6.03	Billie Jean
875	8	Eagles	8.03	Like a Prayer
253	1	The Beatles	8.78	Billie Jean
253	2	Eagles	5.28	With or Without You
929	1	U2	8.48	Smells Like Teen Spirit
929	2	Michael Jackson	6.85	Hey Jude
929	3	Pink Floyd	5.04	Hotel California
929	4	Queen	4.44	Like a Prayer
929	5	Eagles	6.87	With or Without You
929	6	Led Zeppelin	6.82	With or Without You
929	7	Michael Jackson	3.74	Back in Black
929	8	Eagles	9.49	Hey Jude
929	9	Madonna	9.38	Like a Prayer
497	1	U2	7.38	With or Without You
497	2	Michael Jackson	2.7	Back in Black
497	3	Madonna	4.22	With or Without You
497	4	U2	6.82	Bohemian Rhapsody
497	5	The Beatles	9.88	Back in Black
378	1	U2	9.11	Hey Jude
378	2	Queen	3.58	Hey Jude
378	3	U2	5.49	Stairway to Heaven
378	4	The Beatles	4.58	Like a Prayer
378	5	U2	4.49	Stairway to Heaven
497	1	The Beatles	7.82	Like a Prayer
497	2	The Beatles	6.95	Hey Jude

session_id	item_in_session	artist	length	song
4847	1	Mirvana	7.61	Hey Jude
4847	4	Led Zepplin	4.33	With or Without You
4847	5	Michael Jackson	7.87	Like a Prayer
881	1	Madonna	4.32	Stairway to Heaven
881	2	Mirvana	8.61	Stairway to Heaven
889	1	Pink Floyd	5.31	Hey Jude
889	2	The Beatles	5.73	Comfortably Numb
813	1	AC/DC	6.98	Smells Like Teen Spirit
813	2	The Beatles	6.79	Smells Like Teen Spirit
813	3	The Beatles	3.87	Hey Jude
813	4	The Beatles	5.78	With or Without You
813	5	U2	4.51	Back in Black
813	6	Mirvana	9.68	Stairway to Heaven
813	7	AC/DC	5.34	Hotel California
813	8	Mirvana	9.39	Stairway to Heaven
813	9	Eagles	6.54	Bohemian Rhapsody
886	1	Mirvana	2.75	Bohemian Rhapsody
886	2	AC/DC	9.09	Like a Prayer
886	3	The Beatles	4.55	Hotel California
823	1	Michael Jackson	9.77	Hotel California
823	2	Michael Jackson	8.1	Back in Black
823	3	Led Zepplin	5.38	With or Without You
823	4	U2	5.27	Like a Prayer
981	1	Madonna	4.8	Like a Prayer
981	2	The Beatles	8.47	Billie Jean
381	1	Madonna	6.99	Smells Like Teen Spirit
381	2	Queen	5.68	Smells Like Teen Spirit
4444	1	The Beatles	4.68	Smells Like Teen Spirit
4444	2	Mirvana	5.32	Back in Black
4444	3	Eagles	7.47	Stairway to Heaven
4444	4	Led Zepplin	4.28	With or Without You
415	1	Queen	4.8	Bohemian Rhapsody
415	2	Queen	4.34	Hotel California
415	3	Madonna	9.88	Smells Like Teen Spirit
415	4	Eagles	9.07	Billie Jean
415	5	Pink Floyd	5.64	Hotel California
415	6	Mirvana	5.16	Comfortably Numb
415	7	U2	9.81	Billie Jean
415	8	Madonna	3.94	Billie Jean
415	9	Queen	4.07	Hotel California
268	1	Queen	6.82	Back in Black
268	2	Mirvana	8.97	Hey Jude
794	1	The Beatles	8.42	With or Without You
794	2	Eagles	6.42	Like a Prayer
794	3	Led Zepplin	6.59	Hotel California
794	4	Queen	4.41	Comfortably Numb
794	5	Mirvana	9.53	Hey Jude
794	6	Mirvana	9.83	Stairway to Heaven
794	7	Michael Jackson	4.23	Hotel California
523	1	Madonna	3.64	Like a Prayer
523	2	Queen	6.71	Back in Black
738	1	Led Zepplin	7.72	Hotel California
738	2	Queen	7.52	With or Without You
738	3	Mirvana	8.56	Back in Black
738	4	Pink Floyd	7.7	Back in Black
738	5	Michael Jackson	7.82	Hotel California
738	6	Eagles	9.8	Bohemian Rhapsody
738	7	AC/DC	3.04	Billie Jean
738	8	Mirvana	8.66	With or Without You
794	1	Eagles	3.87	Stairway to Heaven
794	2	Mirvana	6.47	With or Without You
794	3	Led Zepplin	2.94	Stairway to Heaven
794	4	Mirvana	3.12	Billie Jean
143	1	Madonna	4.55	Stairway to Heaven
143	2	Queen	5.97	Back in Black
143	3	Mirvana	2.82	Hey Jude
468	1	Eagles	7.97	Smells Like Teen Spirit
468	2	U2	5.83	Bohemian Rhapsody
468	3	Pink Floyd	2.97	Stairway to Heaven
468	4	U2	8.4	Hotel California
468	5	U2	8.1	Billie Jean
468	6	U2	8.64	Hotel California
468	7	Queen	3.27	Stairway to Heaven
468	8	The Beatles	9.99	Bohemian Rhapsody
468	9	Eagles	8.9	Hey Jude
546	1	The Beatles	3.89	With or Without You
546	2	Queen	5.97	Stairway to Heaven
546	3	Eagles	4.43	With or Without You
546	4	Michael Jackson	8.36	Smells Like Teen Spirit
184	1	Mirvana	5.53	Like a Prayer
184	2	The Beatles	4.14	Like a Prayer
184	3	Michael Jackson	3.48	Smells Like Teen Spirit
184	4	Michael Jackson	7.54	Like a Prayer
184	5	Michael Jackson	7.2	Bohemian Rhapsody
184	6	U2	7.99	Bohemian Rhapsody
184	7	Mirvana	6.4	Like a Prayer
184	8	Michael Jackson	3.18	Hey Jude
184	9	Led Zepplin	2.58	Billie Jean
365	1	Pink Floyd	5.87	With or Without You
365	2	AC/DC	5.62	Hey Jude
365	3	Mirvana	8.46	With or Without You
365	4	AC/DC	3.29	Smells Like Teen Spirit
797	1	AC/DC	3.42	Back in Black
797	2	AC/DC	4.47	With or Without You
797	3	The Beatles	2.88	Stairway to Heaven
797	4	AC/DC	9.22	Billie Jean
797	5	Led Zepplin	6.81	Billie Jean
797	6	Mirvana	6.12	Hey Jude
726	1	U2	6.4	Smells Like Teen Spirit
726	2	U2	8.66	Back in Black

Learning Question:

1. Data Modeling: Why do we have 3 separate tables instead of 1 table with JOINS?
Cassandra is a **NoSQL database** and does **not support JOIN operations** like relational databases. Therefore, data is modeled based on the application's query requirements. Three separate tables are created because each table is optimized for a specific query:

- **Table 1:** Retrieve all songs played in a session.
- **Table 2:** Retrieve all songs played by a specific user during a session.
- **Table 3:** Find all users who listened to a particular song.

This approach provides faster query performance and avoids expensive joins.

2. Partition Keys: What would happen if we used `user_id` alone as the partition key for Query 2?

If `user_id` alone were used as the partition key, then all sessions of the same user would be stored in the same partition. When retrieving songs for a specific session, Cassandra would have to scan through all of that user's session data, making the query slower and less efficient.

Using `(user_id, session_id)` as the partition key stores each user's session separately, allowing Cassandra to retrieve only the required session quickly.

3. Clustering Order: Why use ASC for one table and DESC for another?

The clustering order is chosen based on how the data is queried:

- **ASC (Ascending)** is used in `song_info_by_session` so that songs appear in the order they were played during the session.
- **DESC (Descending)** is used in `song_playing_history_by_user` so that the most recently played songs appear first.

This matches the expected output of each query and improves retrieval efficiency.

4. UPSERT Behavior: What happens when you INSERT a row with an existing PRIMARY KEY?

In Cassandra, an **INSERT** operation with an existing **PRIMARY KEY** performs an **UPSERT**. This means the existing row is updated (overwritten) with the new values instead of creating a duplicate row.

For example, inserting a new song with the same `(session_id, item_in_session)` replaces the previous song stored at that primary key.

5. Consistency: What is the difference between CONSISTENCY ONE and CONSISTENCY ALL?

CONSISTENCY ONE	CONSISTENCY ALL
Requires a response from one replica only.	Requires responses from all replicas .
Faster read/write operations.	Slower because every replica must respond.
Higher availability.	Lower availability if any replica is unavailable.
May return slightly older data if replicas are not synchronized.	Ensures the latest data is returned because all replicas must agree.

Conclusion

This project demonstrates the practical implementation of Apache Cassandra for managing large-scale, distributed NoSQL databases. By designing query-specific tables instead of relying on joins, the Sparkify music streaming database efficiently retrieves session-based and user-based music information. The project also provides hands-on experience with Cassandra installation, keyspace creation, table design, CQL commands, data insertion, and query execution. Overall, the exercise highlights the importance of data modeling in Cassandra and shows how NoSQL databases provide scalable and high-performance solutions for big data applications.